

Audit Market Concentration in the EU: Corporate Decision-Making and Quantitative Analysis - Assignment I

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Abstract

This project uses the TRR 266 Template for Reproducible Empirical Accounting Research (TREAT) to provide an infrastructure for open science-oriented empirical projects. Leveraging the Audit Analytics database on Transparency Report data, this repository showcases a reproducible workflow that integrates Python scripts for data preparation, analysis, and visualization. The project replicates and extends findings from the European Commission's 2024 report by analyzing and visualizing Big 4, CR4, and 10KAP audit firms' market shares in the number of public interest entity statutory audits for 2021 across EU countries and at the EU level. The results include a detailed breakdown of market concentration trends, offering insights into the dominance of major audit firms across different jurisdictions. In doing so, the project documents and discusses the research design choices made and the variations between the original and reproduced results. This code base, adapted from TREAT, provides an overview of how the template can be used for this specific project and demonstrates how to structure a reproducible empirical research project.

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1 List of Abbreviations

EC: European Commission

EU: European Union

IDE: Integrated Development Environment

NCA: National Competent Authority

PIE: Public interest entity

TREAT: TRR 266 Template for Reproducible Empirical Accounting Research

UK: United Kingdom

WRDS: Wharton Research Data Services

2 Introduction

The aim of this paper is to illustrate the use of open science tools in empirical accounting research.

The original report by European Commission (2024b) examines trends in the European Union (EU) market for statutory audits of public interest entities (PIEs) from 2019 to 2021, with a focus on market concentration, audit quality risks, and the effectiveness of audit committees. This report is the third check-in on how the EU’s auditing standards and market structures are evolving since the *Article 27 of Regulation (EU) No 537/2014* (2014) was introduced, which introduces a host of specific audit requirements for PIEs. A statutory audit is a legally mandated evaluation of financial records aimed at certifying the accuracy of companies’ financial statements, thereby enhancing stakeholder confidence and supporting the orderly functioning of markets (European Commission 2024a). The report by European Commission (2024b) argues that while statutory audits play a crucial role in fostering confidence in financial markets, persistent market concentration and quality deficiencies highlight the need for strengthened oversight and improved transparency to ensure the resilience and reliability of audit practices across the EU.

This paper replicates Figure 3 from the European Commission’s (2024b) report, which visualizes the market shares of audit firms — Big 4, CR4, and 10KAP — by the number of PIE statutory audits conducted in 2021 across EU countries and at the EU level. The replication provides insights into market concentration and the dominance of major audit firms, reflecting variations in audit practices and competition across Europe.

The project uses data from the Audit Analytics database on Transparency Reports via Wharton Research Data Services (WRDS) for the year 2021, covering PIEs across EU countries. The replication closely follows the methodology outlined by European Commission (2024b). It focuses on three key market concentration measures: the market share of the Big 4 audit firms, the CR4 market share for the largest four firms in each country, and the 10KAP market share, which includes the Big 4 and additional major networks. These measures provide a detailed assessment of audit firm dominance and competition, offering valuable insights into the structure and dynamics of the statutory audit market in Europe.

The analysis involves gathering and filtering Transparency Report data, identifying rele-

vant PIEs and EU countries, and calculating the Big 4, CR4, and 10KAP market shares for each country as well as at the EU level. Explicit assumptions were made whenever the document by European Commission (2024b) was unclear on how to proceed. The entire Python computation code for market share calculations is available in `code/python/do_analysis.py` for detailed review if necessary. This paper focuses on the replication process, presenting visualizations, and discussing the results.

The replicated figure from original report by European Commission (2024b) is shown below, followed by the research design choices and assumptions in Section 3, documentation of the replication steps in Section 4, a detailed results comparison in Section 5, and concluding remarks in Section 6.

Figure 1 illustrates the market shares of audit firms — Big 4, CR4, and 10KAP — for the number of PIE statutory audits conducted in 2021. Countries are sorted in a specific order reflecting the replication of Figure 3 from the European Commission’s 2024 report. The discussion of replication findings is presented in Section 5.

[Figure 1 about here.]

3 Research Design Choices and Assumptions

The aim of Assignment I is to replicate a specific empirical figure (Figure 3) from the report by European Commission (2024b). This figure involves analyzing the market shares of audit firms (Big 4, CR4, and 10KAP) across EU countries and at the EU level for PIE statutory audits conducted in 2021. The replication process includes data loading, preparation, and cleaning, followed by the computation and visualization of market concentration measures. For Assignment I, I pulled data from the Audit Analytics database via WRDS and used the Python programming language to carry out the empirical analysis. Visual Studio Code was used as the Integrated Development Environment (IDE) for writing, debugging, and optimizing the Python code.

The replication is based on data pulled from the Audit Analytics Transparency Reports database, specifically from `audit_europe.feed76_transparency_reports` table, which was filtered and aggregated for the analysis. The table offers Transparency Reports published annually by audit firms based in the EEA and Switzerland, along with the names of all entities listed in each

audit firm’s Transparency Reports (Wharton Research Data Services 2024a).

Following European Commission (2024b), I focus the analysis on statutory audits conducted for the fiscal year 2021, ensuring that the data accurately reflects the audit market structure in the EU as specified in the original report. The replication aims to mirror the research design as closely as possible with the available data.

In addition, I impose the following assumptions to ensure clarity and consistency where the report by European Commission (2024b) does not provide explicit guidance:

1. The original report references the reports prepared by national competent authorities (NCAs) responsible for audit oversight as the main source of data, where the data refers to the years 2020 and 2021 and was collected in 2022 (European Commission 2024b, 1). However, this replication uses data obtained from the Audit Analytics Transparency Reports database via WRDS, which provides the latest available version as of September 2024, updated quarterly (Wharton Research Data Services 2024a). Due to potential adjustments and updates made to the database since the original data collection period, there may be differences that could affect the results. For example, audit firms may disclose updated information in subsequent reports, which would be reflected in the later database version rather than the historical snapshot used in the original report. Furthermore, the data vendor regularly updates its databases to correct errors and add new information, which may be included in the later data but not in the 2022 snapshot.
2. The original paper outlines key terms that will be used in this project to ensure consistency and accuracy in the replication. European Commission (2024b) defines the ‘Big Four’ as Deloitte, EY, KPMG, and PwC; the ‘CR4’ as the four largest audit firms in each country; and the ‘10KAP’ as a group consisting of Baker Tilly, BDO, Deloitte, EY, Grant Thornton, KPMG, Mazars, Nexia, RSM, and PwC.
3. As for the geographical scope, I use the variable `trans_report_auditor_state` to identify the country associated with each audit firm rather than the audited company’s headquarter country (`headquarter_country` variable). Since the objective is to analyze audit market concentration and the market share of audit firms within specific countries, the country of the audit firm’s reported activity is more relevant than the headquarter country of the entities they audit.

4. Another potential source of discrepancies between the original and replicated tables may be the choice of variables pulled from Audit Analytics. For example, in the `audit_europe.feed76.transparency_reports` table, I chose `entity_map_fkey` over `entity_map_key`, as this unique identifier is created by WRDS for merging (Wharton Research Data Services 2024a). Since the report does not specify the choice of variables used from the database, this could cause differences in the results.
5. For the purpose of replicating Figure 3, I use the `report_year` variable as the primary filter to define the relevant data for year 2021. This choice is based on the fact that `report_year` represents the idealized fiscal year of the Transparency Report, based on the *Europe Transparency Reports Data Dictionary* by Wharton Research Data Services (2024b), which aligns directly with the year-based aggregation needed for Figure 3. While the `report_period_end_date` provides the exact end date of each reporting period (Wharton Research Data Services 2024b), using `report_year` allows for a consistent, annual aggregation without requiring additional alignment of various period end dates. This approach simplifies the data extraction and ensures that all entries correspond to the fiscal year 2021.

By following the steps provided in Section 4 and adhering to the assumptions made, I successfully replicated the analysis and produced the required figure. A thorough step-by-step approach, with each step clearly documented, helped to understand and verify the outputs.

4 Replication Steps

4.1 Step 1: Pulling the Data and Managing the Database

This Assignment involves pulling data directly from the Audit Analytics database and preparing the data for further analysis from raw data to final output.

To ensure data relevance, the pulling process was restricted to data for the year 2021. The variable `trans_report_auditor_state` was used to filter the data to include only audit firms associated with the specific EU and EEA countries, as specified in configuration file `config/pull_data_cfg`. This filtering step ensured that the analysis focuses on audit market concentration within the geographical scope and year range as defined by European Commission (2024b).

4.2 Step 2: Data Preparation

The raw Transparency Report dataset initially contained 13,385 observations, representing entries for the year 2021. To ensure consistency with European Commission (2024b), the abbreviation ‘GR’ for Greece was replaced with ‘EL’ in the `trans_report_auditor_state` field, affecting 188 rows.

The `number_of_disclosed_pies` variable provides the clearest indication of statutory audits conducted by each auditor for PIEs. The definition of PIEs varies across EU and EEA countries (Accountancy Europe 2017), but if Transparency Reports consistently follow country-specific definitions, using the provided variable is reasonable. An important verification step involved comparing the reported `number_of_disclosed_pies` with the count of unique `entity_map_fkey` entries for each `transparency_report_fkey`. This analysis revealed that 5,696 transparency reports included fewer entities in the dataset than the disclosed PIEs, suggesting that some PIE entities might be missing from the dataset. This discrepancy reflects missing PIE data rather than the inclusion of non-PIE entities, hence it is assumed that all entities in the dataset are PIEs, maintaining consistency in the framework for analysis.

Additionally, missing values in key fields, including `transparency_report_fkey`, `entity_map_fkey`, `auditor_fkey`, and `trans_report_auditor_state`, were checked, and it was confirmed that no rows needed to be removed due to missing data, ensuring the completeness of the dataset. To focus on relevant data, rows with `number_of_disclosed_pies` equal to or less than zero were filtered, but no such rows were identified, and all observations were retained.

Duplicate entries in the `entity_map_fkey` column were then analyzed, revealing 1,011 duplicate entities. I assume that duplicate entries in the dataset do not introduce significant issues for the replication of Figure 3. Duplicate entries for the same entity are expected in cases where an entity is audited by multiple audit firms or networks, potentially as part of a joint audit. Since these duplicates retain consistent information on `trans_report_auditor_state`, they do not distort country-level distributions. Additionally, while duplicates might expand representation across different auditor networks or groups, the essence of Figure 3 — focusing on aggregate distributions — remains unaffected. Therefore, these duplicates were not removed.

Auditor network names were standardized to ensure consistency, with names like `|Mazars`

Worldwide|Praxity Global Alliance| simplified to |Mazars Worldwide|, for example. This standardization helped avoid double-counting and unify same entries under consistent naming conventions.

Audit firms were then categorized into groups, including the Big 4 (Deloitte, EY, KPMG, and PwC), the 10KAP (which includes the Big 4 along with other major networks listed in Section 3), ‘Unaffiliated’ firms (those outside the Big 4 and 10KAP). The `auditor_network` field contained 636 observations with missing values, which were grouped under the “Other (Blank)” category. I assume that these blank entries do not belong to the Big Four or 10KAP networks but may still contribute to the CR4 category in some countries. So, these entries were retained in the dataset and reviewed during the aggregation stage to validate their relevance.

These preparation steps ensured alignment with the methodology of European Commission (2024b) and the dataset’s readiness for subsequent analysis.

4.3 Step 3: Analysis Implementation and Figure Reproduction

The analysis step computes market concentration metrics for audit firms across EU countries and at the EU level using the prepared transparency data. The analysis begins by calculating the market shares of the Big 4, 10KAP, and CR4 audit firm groups for the number of statutory audits conducted in 2021. For each country, the Big 4 market share is derived as the percentage of PIE statutory audits conducted by Deloitte, EY, KPMG, and PwC, while the CR4 market share represents the top four audit firms in each country based on audit counts. The 10KAP market share includes the Big 4 along with six additional networks, providing a broader view of market concentration.

The following formula is used to calculate market share:

$$\text{Market Share (\%)} = \left(\frac{\text{Number of PIE Audits by Network Group}}{\text{Total PIE Audits in Country}} \right) \times 100$$

The analysis also includes the creation of a bar chart visualizing market shares for all countries and the EU aggregate, saved in both PNG and pickle formats. The aggregated market share data is saved in CSV format, ensuring reproducibility and further analysis opportunities.

5 Results

To verify the results of analysis step and observe trends before plotting, I have additionally saved the CSV output in `aggregated_market_shares.csv`. The analysis of market shares for statutory audits across European countries reveals distinct patterns of market concentration and auditor dominance. Big 4 firms exhibit a significant presence, often commanding a dominant share in many countries. For example, countries such as the Czech Republic, Estonia, and Finland demonstrate Big 4 market shares exceeding 95%, reflecting the significant influence of these firms in specific markets. However, the extent of this dominance varies, with lower Big 4 shares observed in countries like Bulgaria (17.5%) and Greece (43.6%), indicating a more diversified audit market in these regions. The analysis reveals that the Big 4 hold a market share exceeding 80% in 11 EU countries. This aligns with the finding of European Commission (2024b), which similarly highlights that the Big 4 dominate over 80% of the market share in 11 Member States.

The inclusion of additional audit networks through the 10KAP grouping increases the market share substantially across all countries, as expected. This is particularly evident in countries like Ireland, Malta, and the Netherlands, where the 10KAP share reaches 100%, reflecting the full inclusion of statutory audits under this grouping. In contrast, countries like Bulgaria and Poland, with 10KAP shares of 41.8% and 63.6%, respectively, suggest the presence of a substantial number of statutory audits conducted by unaffiliated or smaller firms.

In the analysis of CR4 market shares, I verified whether the four largest audit firms (CR4) in each country overlapped entirely with the Big 4 audit firms. This overlap occurred in 15 out of 28 countries, where CR4 and Big 4 market shares were identical. In the remaining 13 countries, at least one non-Big 4 firm contributed significantly to the statutory audits, resulting in distinct CR4 market shares. These findings align with European Commission (2024b), which notes that in 13 Member States, the Big Four are not the four largest audit firms in terms of the total number of PIE statutory audit opinions. This highlights the gradual diversification in the audit market in some countries, where non-Big 4 firms are playing a more prominent role, while the Big 4 continue to dominate in others.

At the EU level, the aggregated market shares further underscore the dominance of these major players. The Big 4 account for 70% of statutory audits, compared to European Commission

(2024b), which notes an average EU market share of 59% for the Big 4 in 2021, down from 70% in 2018. The broader 10KAP grouping in my output reaches nearly 90%, exceeding the 81% reported by European Commission (2024b). The CR4 market share at the EU level also stands at 70%, aligning with the Big 4, and is consistent with the average CR4 market share reported by European Commission (2024b). These results highlight the varying degrees of market concentration across Europe, with certain countries demonstrating a highly centralized audit market dominated by the Big 4, while others display a more distributed market landscape, incorporating both major and smaller audit firms. The slight discrepancies in the figures suggest potential differences in methodologies, data coverage, or sample definitions.

Based on the results for Austria, the Big 4 market share is 86.7% in my analysis, aligning closely with the approximate 85% represented in bar chart of European Commission (2024b). Similarly, the 10KAP market share of 92.55% in my analysis corresponds well with the slightly higher bar in the report, and the CR4 market share of 86.7% aligns perfectly with the Big 4 share, confirming the dominance of these firms in Austria’s audit market. This consistency highlights the robustness of the replicated analysis for this country.

The results for Romania show notable discrepancies. The EC Report indicates market shares of 15% for the Big 4, 16% for CR4, and 24% for 10KAP, while my analysis reports significantly higher values: 63%, 70.61%, and 78.38%, respectively. These differences likely arise from assumptions made in Section 3 like variations in PIE definitions and auditor classifications. The report by European Commission (2024b) may use stricter criteria, reflecting Romania’s decentralized market where non-Big 4 firms dominate, unlike the broader inclusion in my dataset.

In summary, the replication effectively captures the original report’s key findings, confirming the dominance of the Big 4 and the broader 10KAP group in the European audit market. The analysis highlights varying degrees of market concentration across countries, with some demonstrating a more decentralized audit landscape. While discrepancies in individual country results, such as Romania, underline the importance of methodological alignment, the strong similarity between the overall market shares reported in the original report and the replicated results underscores the robustness and reliability of the findings.

6 Conclusion

The project effectively demonstrates the use of a systematic and collaborative workflow for analyzing audit market concentration across Europe, leveraging the TRR 266 Template for Reproducible Empirical Accounting Research. By following a structured approach, I successfully replicated a key figure and empirical findings of European Commission (2024b), offering insights into the dominance of the Big 4 and the broader 10KAP group in the European audit market.

This assignment is an exact replication of the report by European Commission (2024b), where the goal was to closely follow the report's approach wherever possible and make justified assumptions when necessary. While exact replications often yield different samples and outcomes due to dataset updates or methodological variations, my results align closely with the statistics presented in the original study, reinforcing the reliability of the replication process.

The analysis reveals significant variation in market concentration across countries, with the Big 4 accounting for over 80% of market share in 11 countries, while others show a more decentralized landscape where non-Big 4 firms contribute substantially. These findings align closely with the original study results, confirming the reliability of the data and methodologies used in this replication. However, discrepancies in specific countries, such as Romania, highlight the importance of transparent methodologies and consistent definitions when conducting such analyses.

This assignment required a comprehensive application of skills learned throughout the course, incorporating data analysis, replication techniques, and visualization. In the future, this repository can be cloned or forked (if made public) to kickstart further projects on earnings management measures analysis. Thanks for reading!

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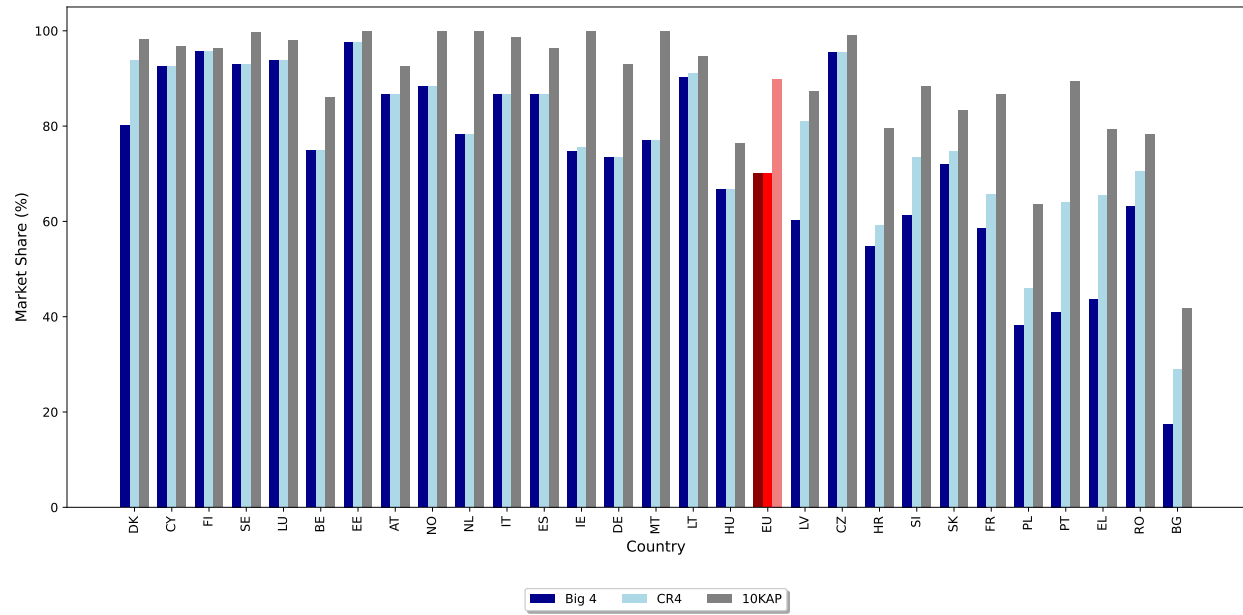


Figure 1: Replicated Figure 3: Audit Firms' Market Shares in the EU for 2021